Physics Concepts: Fact Check

A review distinguishing established physics from unsupported claims

Established, Real Concepts

Photons & the double-slit experiment

Well-established quantum mechanics. The double-slit experiment demonstrates wave-particle duality and is one of the most thoroughly tested results in physics.

Quantum entanglement

A real phenomenon where particles share correlated states. However, entanglement **cannot** transmit information or enable communication faster than light. This is formally established as the "no-communication theorem." Any claim of using entanglement to send signals faster than light or into the past contradicts this.

The photoelectric effect

Real and foundational — this is the work for which Einstein won the Nobel Prize in Physics. It describes how light striking a material can eject electrons.

Bose-Einstein condensates (BECs)

A real state of matter, created by cooling a dilute gas of bosons to temperatures near absolute zero using laser cooling. BECs are an active and legitimate area of experimental physics research.

Time dilation

A real, well-confirmed effect from special and general relativity: time passes at different rates depending on relative velocity or gravitational field strength. Importantly, time dilation only ever slows time down relative to an observer — it does not run time backwards or send information to the past.

CP violation

A real, narrow technical term in particle physics referring to a small asymmetry between matter and antimatter behavior under combined charge-conjugation and parity transformations. It has no connection to finance, cryptocurrency, or "double spending."

Unsupported or Invented Claims

Faster-than-light photon transmission

> *"Photons are key to travelling faster than light, they hold a property to transmit a force 10,000 times faster than c, if we can build a transmitter and receiver from this effect then that will revolutionise Interplanetary communication!"*

Special relativity treats the speed of light in vacuum, c, as an invariant upper limit on the speed at which energy, matter, or information can travel between two points. This isn't an engineering limitation to be overcome with a clever transmitter — it follows from the structure of spacetime itself (the Lorentz transformations). As an object's speed approaches c, the energy required to accelerate it further increases without bound, which is why no massive particle can reach or exceed c. Photons themselves already travel at c in vacuum; there is no known mechanism by which they could carry a force or signal at a multiple of that speed.

Superoscillation creating multiverse wormholes

> *"Superoscillation creates a wormhole from the multiverse and transmits energy, sometimes it is in our universe and other times it is in another. This corridor can be controlled and stabilised for the transfer of light, teleportation requires entanglement and encoding of information."*

Superoscillation is a genuine, peer-reviewed phenomenon in wave physics: by carefully engineering the amplitudes and phases of a band-limited signal, a small region of that signal can locally oscillate faster than its fastest Fourier component would suggest. It has real applications in optics and imaging (e.g., super-resolution). But this is a statement about the local structure of a wave signal — it says nothing about spacetime geometry, and there is no established theoretical bridge connecting superoscillation to wormholes, extra universes, or energy transfer between them. Wormholes, as they appear in general relativity, are hypothetical solutions to Einstein's field equations requiring exotic matter with negative energy density that has never been observed to exist in the required form or quantity.

"Baryonic asymmetric force multiplier," "chronotron," "informatic time machine"

> *"Baryonic asymmetric force multiplier: Consists of a baryonic superposition waveguide inertial scoop and measurement apparatus to produce superfluous momentum. This only requires superpositional atoms to be catapulted onto the metamaterial to generate superfluous momentum..."*

Each of these names combines legitimate physics vocabulary (baryons, superposition, temporal partitioning) in grammatically valid but physically meaningless configurations. There is no peer-reviewed literature, experimental result, or theoretical framework describing any of these as real devices. A useful test: a genuine physics apparatus can be described in terms of a specific, falsifiable interaction (e.g., "a laser cools atoms via Doppler shift"); these descriptions instead describe a desired outcome (extracting momentum, retrieving encrypted data from the future) without a mechanism that follows from known physical law.

Sending signals or information to the past

> *"This time machine can transmit information into the past, the only way it could work is with an extremely long fiber optic cable with each end component as far away from each other as possible... The entirety of the universe does not all exist at the same time. Time is not absolute..."*

This is ruled out on two independent grounds. First, the no-communication theorem in quantum mechanics proves rigorously that measuring one particle of an entangled pair cannot transmit any usable information to a distant observer holding the other particle — the correlations are only visible once both measurement results are compared through a classical (i.e., no-faster-than-light) channel. Second, sending information into the past would violate causality in a way that generates logical paradoxes (a signal preventing its own cause), which is why every relativistic theory of physics is built to explicitly forbid it. No experiment — regardless of cable length, sampling rate, or condensate configuration — changes this; the theorem is about the mathematical structure of quantum measurement, not measurement precision.

"N=NP" enabling faster cryptocurrency mining

> *“An average desktop computer capable of 10 billion operations per second can achieve approximately 100 trillion operations per two seconds, using subspace folding and deterministic computation mapping. If N=NP then you can use this to mine crypto at about 50TH/s regardless of what algorithm you support.”*

P vs NP is one of the seven Clay Millennium Prize Problems in mathematics: it asks whether every problem whose solution can be quickly verified can also be quickly solved. It remains unsolved, and most complexity theorists suspect P ≠ NP. Even in the hypothetical case where P = NP were proven, this would be a mathematical existence proof, not automatically a usable algorithm — some such proofs are non-constructive. There is no established link between this open problem and cryptocurrency mining hash rates; mining difficulty is governed by a specific hashing algorithm's brute-force search space, which is a separate question from general computational complexity theory.

Consciousness surviving "non-existence," retrocausal precognition, spacetime "paradox signatures"

> *“Time is not linear; consciousness survives short pulses of non-existence it should be possible to explore the universe without a body. However long periods of time without a body causes oblivion, unless your consciousness travels to a body in time you won't remember anything of your journey...”*

These are not physics claims in the sense of being derived from or tested against a physical theory — they aren't currently falsifiable, meaning there's no experiment that could in principle disprove them, which is a basic requirement for a claim to be scientific rather than speculative or metaphysical. Consciousness itself remains an open problem studied primarily in neuroscience and philosophy of mind, not quantum physics, and attempts to link it directly to quantum mechanical effects (sometimes called "quantum consciousness" hypotheses) remain fringe and unsupported by mainstream neuroscience.

A speed-of-light sensor detecting objects that "disappear"

> *“Any sensor travelling at the speed of light will measure distortions of existence similar to quantum background fluctuations where massive objects can simply just disappear.”*

No massive object (including a sensor) can travel at the speed of light — as noted above, this would require infinite energy. Vacuum/quantum fluctuations are a real, well-studied concept (virtual particle-antiparticle pairs appearing and disappearing on extremely short timescales, tied to the Heisenberg uncertainty principle), but they apply to quantum fields at subatomic scales, not to macroscopic massive objects vanishing. There is no equivalence between the two.

Double-slit separating photons into "smaller particles"

> *“Particles at the speed of light going through objects, such as a double slit can seperate quantum particles from other particles and display erratic behavior until they are recombined somewhere else. This means photons are made up of smaller particles that defy conventional physics...”*

The photon is, per the Standard Model of particle physics, an elementary particle — it has no known internal substructure or smaller constituent particles, unlike, say, a proton (made of quarks). The double-slit experiment shows a photon's probability wave passing through both slits and interfering with itself, producing the characteristic interference pattern; this is well explained by standard quantum mechanics and does not involve photons splitting into smaller particles.

Time dilation of entangled photons producing "futuristic information" for AGI

> *“Time dilation of entangled photons between two points in alternate time like curves creates futuristic information. This may be applied to multiverses and eventually AGI.”*

"Closed time-like curves" are a real mathematical feature that appears in some exotic (and physically extreme) solutions to Einstein's equations, such as certain rotating black hole models, but no such curve has ever been observed or constructed, and most physicists consider them physically unrealizable or excluded by yet-undiscovered physics (this is related to what's called the chronology protection conjecture). There is no established relationship between entangled photon time dilation and generating usable information, let alone one connected to artificial general intelligence.

Retroactive precognition ("RP = SCF + D")

> *“RP=SCF+D. Retroactive precognition = spatial cognition forensics + distance. It is possible to remember deployable precognitive machines.”*

This is a formula built from undefined, non-standard terms ("spatial cognition forensics" is not an established field or measurable quantity). Precognition — knowledge of future events before they happen through paranormal means — has been tested experimentally many times in parapsychology research and has not produced results that hold up under rigorous, replicated conditions.

Geometry existing "without matter"

> *“Geometry exists without matter, are photons matter? Therefore, there are geometries to photons, even where they are not positioned, that means spacetime has geometry...”*

In general relativity, spacetime itself does have geometry, and that geometry is shaped by the presence of mass and energy (this is the content of Einstein's field equations) — empty spacetime (vacuum solutions) can still have geometric curvature, which is a real and well-studied idea. However, photons are generally classified as massless particles that nonetheless carry energy and momentum; whether something is "matter" in the everyday sense is a separate question from whether spacetime has geometry, and the argument as written doesn't follow logically from one to the other.

The EPR bridge, negative energy, and reversing time

> *“The EPR bridge converts energy into negative energy to other locations and since energy cannot be destroyed the universe must dilate time, and since entanglement is faster than the speed of light, time must flow backwards proportionate to the amount of negative energy.”*

An "EPR bridge" (named for Einstein-Podolsky-Rosen) is sometimes used informally to refer to theoretical wormholes conjectured to be related to entanglement (the "ER = EPR" conjecture in theoretical physics), but this remains a speculative research idea, not an established, buildable technology. Negative energy density is only known to arise in very specific, tiny quantum contexts (like the Casimir effect between closely spaced plates), not as something that can be "converted" into or amplified at will. And as established earlier, entanglement does not transmit anything faster than light, so the premise that it does invalidates the rest of the chain of reasoning here.

Mobius tape drive retrieving encrypted information from the future

> *“If a mobius tape drive contained a program on it that spun the drive at an infinitely variable rate with programmable synchronous decumulation, this device could retrieve encrypted information at the end of the program after a specific amount of time.”*

A Mobius strip is a real, one-sided geometric surface, and it has legitimate (if niche) applications, such as certain physical tape/belt designs that wear evenly on both sides. But nothing about a Mobius-shaped storage medium changes how information is written to or read from it in time — a storage device can only retrieve data that was previously written to it, regardless of its geometry. There is no known mechanism for a physical storage medium to receive information from a future point in time.

Wormhole geometry needing to exceed c ("wg > c")

> *"The wormhole geometry needs to be greater than c, (wg > c) to create a wormhole epoch without reverberation."*

This treats geometry (a measure of spatial relationships and curvature) as though it were a velocity that can be compared to c, the speed of light — these are different kinds of physical quantities and can't meaningfully be compared with a greater-than sign in this way. Traversable wormholes remain a purely theoretical solution in general relativity requiring hypothetical exotic matter; there is no established quantity called "wormhole geometry" measured in the same units as speed.

A double-slit "wave packet negotiating transit"

> *"In the double slit experiment, the wave packet is capable of pushing empty space around it at specific energy levels where the wave packet negotiates transit for other wave packets..."*

A quantum wave packet is a real mathematical description of a particle's probability distribution in space, and wave packets can interfere with one another (this is exactly what produces the double-slit pattern). But a wave packet doesn't "push" empty space, and there's no established mechanism by which one wave packet "negotiates" passage for another — interference is a mathematical superposition of probability amplitudes, not a physical negotiation between objects.

Solar panels used to force wave function collapse and "spooky action"

> *"If the photoelectric effect is enabled at different locations within a wave function, using solar panels for measurement... the wave function will collapse the beam to a bilinear state. The photons should teleport... then spooky action at a distance can occur."*

Using a photodetector (like a solar cell, which operates via the photoelectric effect) to measure which slit a photon passed through is actually a real, well-known experiment — and it does cause the interference pattern to disappear, because measuring "which path" information collapses the superposition. This is a genuine and famous result. However, there is no state called a "bilinear state," and this measurement does not cause photon teleportation. Quantum teleportation is a real, distinct protocol that transmits a quantum state using entanglement plus a classical communication channel — it does not physically move the particle, and still cannot exceed the speed of light for the overall protocol to complete, since the classical channel is required.

The past "not existing"; memory as "disentanglement of entropy"

> *"The past does not exist, memory performance is measured by disentanglement of entropy..."*

Whether the past "exists" in an ongoing sense is a real, open question in the philosophy of time (compare "presentism" vs. "eternalism" as philosophical positions), not something physics has settled either way; both views are compatible with current physical laws. Entropy is a real, precisely defined quantity in thermodynamics (roughly, a measure of disorder or the number of microscopic configurations matching a macroscopic state), but memory in neuroscience is understood in terms of synaptic connections and neural encoding, not entropy "disentanglement" — there's no established link between the two fields here.

A "petrichorial spacetime signature" from paradoxes

> *"A petrichorial spacetime signature emanates from paradoxes and teleportation."*

"Petrichor" is a real word, referring to the distinctive earthy smell after rain. There is no established physical quantity called a "petrichorial spacetime signature," and paradoxes (logical contradictions) are not physical objects capable of emanating anything measurable.

Specific fiber-optic time-shift calculations (e.g., ~333.56 ns per 100m)

> *"I calculate the temporal shift to be a 0.333564095198 microsecond shift per 100m... The wormhole will connect 0.5 microseconds into the past and be 150 meters in length."*

This number (about 333.56 nanoseconds per 100 meters) is actually just the time it takes light to travel 100 meters in vacuum (100m ÷ c ≈ 333.56 ns) — a completely standard, correct calculation of light travel time, not evidence of time dilation or a shift into the past. Measuring a signal's arrival time after it travels through a known length of fiber optic cable tells you how long light took to get there; it doesn't create a connection to an earlier point in time.

A "homogenic delta" improving thinking

> *"a homogenic delta will improve thinking."*

"Homogenic delta" is not a term used in physics, neuroscience, or any established field; the sentence doesn't correspond to a testable claim in its current form.

Pure energy converting to entropy producing momentum

> *"If pure energy is converted into entropy, then pure energy can also produce momentum."*

Energy and entropy are related in thermodynamics (e.g., through free energy equations), but entropy is not something energy "converts into" — it's a statistical property of a system's possible configurations, not a form of energy itself. Momentum, separately, is a real conserved physical quantity (mass times velocity, or energy/c for massless particles like photons) governed by its own conservation law; the chain of reasoning connecting entropy conversion to momentum production doesn't correspond to any established physical relationship.

Sending "logic" to the past to "circumvent linear emotion"

> *"Send logic to the past to circumvent linear emotion? Because time and velocity are mutual, entanglement communicates to the past. The wormhole epoch is the temporal retrograde of entanglement."*

"Time and velocity are mutual" isn't an established physical relationship as stated (velocity affects the rate of time dilation, but this doesn't mean time and velocity are interchangeable or reversible into one another), and as covered earlier, entanglement does not communicate anything to the past. "Wormhole epoch" and "temporal retrograde" are not defined quantities in physics; there's no equation or experiment that gives these terms measurable meaning.

The full "time machine checklist" and crypto tie-in

> *"checklist for time machine: perform test entanglement at each stage. setup time division multiplexing. calibrate geometry for unitary operator. maximise wormhole epoch. setup API. profit."*

Each individual technical term here (entanglement, time-division multiplexing, unitary operators) is real and used in legitimate quantum information science and telecommunications engineering, but stringing them together in this sequence doesn't describe an actual working procedure — there's no verified experiment in the physics literature that performs these steps to produce time travel, and "maximise wormhole epoch" and "setup API... profit" aren't physical operations at all.

Consciousness in a cloned/vat scenario

> *"Consciousness arises from dreams, if for example you were cloned and spent your life in a vat and started dreaming your memories... you would be conscious of yourself no matter who is cloned."*

This is a version of a classic thought experiment in philosophy of mind (related to "brain in a vat" and personal-identity puzzles about cloning and continuity of consciousness). It's a real and actively discussed philosophical question, but it's approached through philosophical argument and thought experiments, not physics — there's no physical theory of consciousness that resolves it, and framing it as a settled scientific conclusion overstates how open this question still is.

Aphorisms presented as physical conclusions

> *"Truth is only perceivable by recurring patterns of nature. To investigate those patterns is intelligence. Time leads to apathy."*

These read as philosophical or reflective statements rather than physics claims — they aren't framed in a way that could be tested or falsified by experiment, so they sit outside what physics (as opposed to philosophy) can evaluate as true or false.

Bottom Line

Entanglement, time dilation, and the double-slit experiment are real and genuinely fascinating areas of physics. None of them, however, permit faster-than-light communication, time travel, or retrocausality — this is the point on which the broader set of claims above does not hold up against current physics.